A Crash Guide on Ethics for AI Developers

problems, perspectives, and directions for further reading

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How to use this text

This short handbook serves as a first introduction to ethics and to ethical problems that you will encounter (as I did) as an AI designer or developer. It introduces concepts, examples, and perspectives that help make those problems visible. It does ***not*** *offer solutions, a complete survey of ethics, nor a checklist that guarantees ethical AI. The best way to approach this text is as an invitation to investigate further questions relevant to broadly defined ethical issues, because ultimately, as a developer,* ***defending your agency requires a deeper understanding of what is at stake with what you do****. Each section offers reflection pathways. Consider these sources as starting points, not cumulative assignments; they also do not (and need not) necessarily agree with one another. Use them selectively to pursue the questions that interest you most.*

1 Why ethics for AI developers?

For a number of reasons, students entering computational tracks at the university level (such as Computer Science, AI, Information Science, and related fields)—and consequently becoming tech researchers, experts, and leaders in technology—may not have been explicitly exposed to topics broadly related to ethics. The secularization of our societies and a (plausibly inevitable) widespread attitude towards de-politicization in STEM contexts do not provide students with a context in which they can learn to form a position about ethical concerns at large.[1] Unfortunately, this implicit educational selection mechanism creates the risk of naive, if not dangerous, misunderstandings. *A system performs well: does that entail we should use it? A pattern appears in the data: does that mean we should act upon it?* Simple as they may seem, these questions quickly become complex when we broaden our scope of reflection.

Fortunately, this is not "rocket science". These concepts are accessible to all of us simply because we are humans—arguably, the living beings on Earth exhibiting most social complexity. Fellow humans have engaged with these discussions in the past, and will continue to do so in the future. In this sense, becoming acquainted with ethical concerns increases, at individual and at collective levels, our "human capital": our capacity to recognize problems, discuss alternatives with peer humans, and reflect on our choices.

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[1] Politics is not merely a matter of gaining and maintaining power; it is also a matter of deciding how and *for what* that power is sought. The suppression of ethics-oriented discussions becomes collateral damage of depoliticization, plausibly because ethics provides general instruments to evaluate these decisions.

1.1 Technology in the (bad) news

It is not difficult to find news reporting on technical research and deployments that trigger strong negative reactions from the public. For instance, years ago research was reported to infer whether people were gay or straight from a photograph.[2] The COMPAS system, actually used in US criminal justice for predicting recidivism, became the subject of an intense debate on racial bias.[3] The Dutch SyRI system, linking personal data across government agencies to produce welfare-fraud risk alerts, was found to disproportionately impact families with minority backgrounds and raising critical questions about discrimination and lawfulness.[4] Retail surveillance has also become widespread: facial recognition to identify potential shoplifters, pervasive cameras, and increasingly intensive checks at self-checkouts, purportedly resulting in discriminatory outcomes.[5] Similar technological changes are occurring in border control. More recently, chatbot interactions have been associated with suicide, addiction, and triggering of psychotic symptoms, in some cases sparking lawsuits by families and raising distinct questions about foreseeable harm, harmful relationships with technology, and legal responsibility.[6] Although these reports and allegations should not be confused with established findings about causation or liability, it is undeniable that these cases draw strong public reactions across research, technical, legal, and political dimensions.

Then there is a grey area of technological research in the "(possibly, not yet) bad?" news. For instance, research in 2024 utilized detailed sequences of life events to predict aspects of human lives using a *life2vec* model, constructed in analogy to the *word2vec* representation.[7] Other research worked instead on predicting judicial decisions[8]—a capability which may be tied to identifying the underlying, potentially implicit, political leanings of the judge under observation. Studies of this kind offer a clear illustration of how scientific interest in predicting plausible "completions" does not, on its own, establish which downstream applications are ethically acceptable (in these cases, *is it justified to [be able to] predict an individual's mortality, personality, or financial trajectory, or to audit a judge's ruling prior to deliberation?*).

New technologies can also bring structural socio-economic disruptions that may seem unavoidable. Decades ago, the digitization of audio and video through MP3 and MPEG-4 formats, alongside peer-to-peer (P2P) networks, made digital piracy commonplace until streaming platforms established a new legal economic model (although the vast majority of artists arguably saw their income decreasing along with this transition). Today, with the current generative AI wave, new tools like automated reverse-engineering coding agents make it remarkably easy to reconstruct expensive proprietary software. This possibility fundamentally alters the competitive leverage traditionally held by established software companies, and thus transforming the entire economic pipeline of software engineering. Widespread job losses may become a direct consequence of the absence of strategic organizational and societal rearrangements. Scaling similar dynamics across the broader tertiary sector (so called white-collar professions)—and potentially secondary (blue-collar) and primary sectors through e.g. robotics and precision agriculture—makes it clear

[2] Sam Levin, "New AI can guess whether you're gay or straight from a photograph", *The Guardian*, 7 September 2017.

[3] Julia Angwin, Jeff Larson, Surya Mattu, and Lauren Kirchner, "Machine Bias: There's software used across the country to predict future criminals. And it's biased against blacks", *ProPublica*, 23 May 2016.

[4] Public Interest Litigation Project, "Court hearing against risk profiling of Dutch citizens by SyRI", *PILP*, 29 October 2019.

[5] Rowland Manthorpe, "Co-op is using facial recognition tech to scan and track shoppers", *Wired UK*, 3 December 2020; NL Times, "Jumbo takes extra measures against self-scan shoplifting," *NL Times*, 11 February 2024.

[6] Eileen Guo, "An AI chatbot told a user how to kill himself — but the company doesn't want to 'censor' it," *MIT Technology Review*, 6 February 2025; Nadine Yousif, "Parents of teenager who took his own life sue OpenAI," *BBC News*, 27 August 2025.

[7] Germans Savcisens et al. "Using sequences of life-events to predict human lives". In: *Nature Computational Science* 4 (2024), pp. 43–56.

[8] Daniel Martin Katz, Michael J. Bommarito II, and Josh Blackman. "A general approach for predicting the behavior of the Supreme Court of the United States". In: *PLOS ONE* 12.4 (Apr. 2017), pp. 1–18. URL: https://doi.org/10.1371/journal.pone.0174698.

that a collective discussion on how to reshape our societies is urgently needed.[9]

All these cases and scenarios carry their own distinct problematic core (e.g., algorithmic bias and discrimination, dual-use or *mission-creep* implications, or macro-economic disruption), rather than being simple illustrations of a single universal problem. The common lesson at this point is modest, yet crucial: *technical performance does not, on its own, establish ethical justification for its deployment, sometimes even for its introduction.*

Exercise 1. Purposes of use, stakeholders and outcomes.

Select two or three of the technical systems discussed above and answer the following questions:

- *What is the system supposed to achieve? Is there a discrepancy between the purpose intended by its designers and the goals of its (potential) users?*
- *Identify the main stakeholders. Then consider two deployment scenarios—one producing mostly positive outcomes, the other mostly negative. Who actually bears or encounters these (potential) consequences? Are there stakeholders which bear a qualitatively opposite outcome?*
- *What was hidden by the initial performance/optimization metric?*

2 Morals & co.

Etymologically, the word "morals" derives from the Latin *mores*, referring to the social norms, customs, and habits of a community. The equivalent term in ancient Greek, *ethos*, is similarly associated with general character, customary ways of living, and shared attitudes. Before entering into ethics, it is meaningful to go deeper into the general functioning of morals.

2.1 Descriptive vs Prescriptive

By definition, individuals behave according to—or against—*mores*. But there is a fundamental difference between describing what people actually do and prescribing what they should do. Two distinct types of norms can therefore be defined:

- **Descriptive norms** concern what is usual, frequent, or expected—in other words, *normal* in a social context. They require both a target of observation (a specific group) and an implicit vantage point (including the observer's perspective and various selection mechanisms, e.g. *world-*, *representation-*, or *aggregation-dependent*).
- **Prescriptive norms** concern what is considered correct or incorrect behavior according to a certain normative source, which may be community standards. Here, too, lies an implicit point of observation: prescribed rules are about what was deemed relevant by whoever established the norm.

The distinction between descriptive and prescriptive (or regulative) norms is crucial; it can be thought of as rooted in the fundamental divide between *modelling/interpreting* and *designing*. Legal scholars explicitly divide their research questions along two main fronts (descriptive: *what the law is*, normative: *what the law ought to be*), while contemporary research in ethics—traditionally regulative in nature—shows a growing interest in empirical and descriptive ethics. Similarly, sociologists and anthropologists are often distinguished in social sciences by whether they aim to intervene in social and organizational processes (e.g., action research) or investigate them with minimal intervention and mediated reflection (e.g., ethnography and ethnology).

[9] A growing body of literature shows that Luddites, a bottom-up movement of people destroying textile machinery during the beginning of the Industrial Revolution, were not opposed to machines in themselves, but to the devaluation of worker autonomy and the erosion of their worth to society.

A parallel divide exists for software and AI developers. As engineers, we operate naturally in the domain of *prescriptive norms*—we design control systems, including in the form of algorithms and data models to bring about desired outcomes. As data scientists, in contrast, we operate in the domain of *descriptive norms*—extracting patterns and statistics from observed data.

2.2 Abnormal vs "Pathological"

Evidently, borrowing a perspective from medicine, being *abnormal* (i.e., not conforming to descriptive norms) does not automatically entail being *pathological* (i.e., violating "prescriptive" or functional norms). For instance, historically, research in medical imagery recognized that variations in biological traits (such as e.g. the size or shape features of kidneys) across global populations are rarely captured by precise normal distributions; even where bell curves appear, they may simply be due to sub-population majority effects. This explains why, before the advent of deep learning, structural and topological methods were more common than statistical methods in that sub-field. Crucially, even a major structural difference is not necessarily tied to physiological dysfunction. In extreme cases, individuals have been found to live healthy lives with their heart situated on the right side of their chest, or with fluid occupying half of their head. For data scientists, the lesson is clear: identifying a statistical outlier in a dataset tells us merely that a combination of features is rare, but it does not, on its own, justify treating that instance as defective, risky, or illegitimate. In open worlds (and domains depending on social constructions are open), *anomaly detection is not a diagnostic device.*

2.3 In between descriptive and prescriptive

The interaction between descriptive and prescriptive domains can, however, quickly become complex. First, describing the fact that a community recognizes an obligation (even one its members frequently violate) does not mean the observer endorses that norm. Acceptance (in the sense of rule-following) and endorsement (in the sense of agreeing with the validity of the rule) add distinct dimensions for the agent expressing the norm beyond mere description. Second, descriptive norms can turn prescriptive under specific conditions — primarily when used for *identitarian* boundaries to demarcate who belongs to a group and who does not. A typical style of dress, for example, can become a requirement that a "real" member of a subculture is expected to adopt. From this structural behavioral standpoint—as far as my own experience goes—there is no fundamental difference in how punks and financial traders recognize one another.

2.4 Whose *mores*?

Individuals behave according to or against *mores*, which are defined with respect to a community. In other words, behavior is *moral* if it is justifiable according to a moral standard set by that community. Consequently, any phenomenon related to *morality* requires the presence of an evaluative framework; and all moral judgment is rendered from a collective standpoint (the one associated with the mores).

Yet, while *mores* are defined collectively, similar dynamics operate at the individual level. This is not only because we may "contain multitudes", in the words of Walt Whitman, but simply because our present conduct is evaluated against our own past patterns of behavior that have proven acceptable or good:

> *"I have my habits, and my own ways of evaluating behavior as good."*

Conversely, organizations and communities often function as collective agents, seen externally as single entities acting for a certain purpose, and sets standards (prescriptive norms), or at least expectations (descriptive norms) of behaviour:

> *"In our organization, this is how we do things."*

These individual and collective levels interact continuously, but should not be silently collapsed. A personal commitment is not necessarily a community norm; a publicly declared standard (e.g. law) may fail to become a community standard; a community standard is rarely acknowledged, understood, endorsed, or accepted unanimously by all its members.

2.5 Normality vs. Normativity

Following the relations highlighted at the end of the previous section, descriptive and prescriptive evaluative frameworks (*is* vs. *ought*) can also be interpreted in agentive terms (*belief* vs. *desire*), and a collective agent perspective can be added to the individual one (*we* vs. *I*).

Perspective	**Authoritative source**	**Collective orientation**	**Individual orientation**
Normality	Something **is**—or is not—the case	**We believe** it to be—or not to be—the case	**I believe** it to be—or not to be—the case
Normativity	Something **ought**—or ought not—to be the case	**We desire** it to be—or not to be—the case	**I desire** it to be—or not to be—the case

This distinction makes it possible to capture several potential mismatches. For instance, agents may desire something that they do not think ought to happen, or recognize an obligation that they do not desire to fulfill. Science generally operates at the level of normality regarding the world; nevertheless, philosophy of science has shown that what is considered factual is always expressed through historically contingent conceptual paradigms. Thus, a collective orientation component is always present, even in case of objective truths. Whether the ought can be truly objective can be diplomatically deemed out of scope, provided we still agree that, collectively, this is what we ought to do.

Exercise 2. The shop owner and the red tie

Suppose you are a shop owner facing the following scenario: (1) You do not want fraudsters in your shop; (2) Fraudsters typically wear a red tie. You therefore decide (3) not to admit people with a red tie. The first statement expresses a practical preference. The second describes an alleged association. The third turns that association into a behavioural rule for treating people.

- *Is your decision good? Is it right? For whom?*

3 The relation between ethics and morals

In philosophy, *ethics* emerged as a specialized domain of inquiry alongside branches such as *ontology*, *epistemology*, and *aesthetics*. This structural division is not unique to Western thought; analogous traditions exist, for instance, in Indian and Chinese philosophy. Ancient Greek ethical schools—such as the *Stoics*, *Epicureans*, and *Skeptics*—offered distinct guidelines on how one ought to live, including frameworks to examine what distinguishes "virtuous" behavior from "vicious" behavior. What was evident even in antiquity is that none of these schools emerged as a definitive "winner" that settled ethical questions once and for all. Rather, they gathered streams of reflection through which individuals entered into scrutinizing and forming their own moral positions in the world.[10]

This historical view makes clear that ethical discussions operate at a meta-level: rather than merely defining *good behavior* by applying a given evaluative framework, they investigate what constitutes a *good evaluative framework*. Notably, while Anglophone academic traditions often

[10]See for instance Marcus Aurelius and his *Meditations*: one can continue to think and live as a Stoic even after becoming a Roman emperor.

use "moral" and "ethical" interchangeably, Latin-oriented and continental traditions maintain a sharper distinction between the two.

For our purposes, we follow the latter tradition, drawing a clear line between *moral* and *legal* inquiries—concerned respectively with the evaluative standards (*mores*) and the laws already established within a community—and *ethical* inquiries, concerned with the critical examination of the reasons for accepting, modifying, or rejecting those standards or laws.[11] One may behave ethically even while acting against morals or rules provided at collective levels (e.g. whistleblowing against institutional corruption). Conversely, conformity does not, by itself, establish ethical justification (e.g. participating in mob violence).

The existence of distinct ethical frameworks whose conclusions may be conflicting (as it was the case with the ancient ethical schools) does not entail that we should just go with *moral relativism* (or one of its various inflections, such as morality being entirely subjective, or even, more radically, that anything goes). The function of ethics, as a meta-normative discourse, is precisely to offer a space in which these competing arguments are analyzed, weighed, and balanced.[12] Indeed, the primary role of ethicists and jurists is to discuss and deliberate—under given circumstances and with explicit reasons—which normative preferences are most defensible when establishing social *mores* or when interpreting/formulating laws.

Exercise 3. Aggregating moral preferences

Suppose we collect global preferences over moral dilemmas and aggregate them into a decision-making procedure.

- *Is the resulting decision-making "moral"? Is it "ethical"?*

The previous is not an abstract exercise. The "Moral Machine" experiment conducted at MIT years ago collected global preferences regarding several variations of the Trolley Problem.[13] It was conceived while autonomous vehicles started to be tested on the road. The underlying question was: how should such artificial devices act when confronting a moral dilemma? The findings revealed significant variations across countries. Therefore:

- *Is it justifiable that a system applies different moral standards depending on the country one lives in?*

3.1 Where does morality come from?

The previous exercise can be used to motivate an exploration into the sources of morality.

"Natural" morals The simplest way to view morality is as a given set of acceptable and unacceptable behaviors. Traditional accounts of morality appeal to eternal truths expressed by religions, or natural law, but one could add to this list innate moral judgments observed early in child development. Interestingly, reading judgments from ancient legal systems one may find that they bear much resemblance to modern law and seem still sensible nowadays. This universality may suggest that fundamental human cognitive mechanisms may be related, at least in part, to moral judgments. Yet, what support these judgments may change depending on time and age.

[11] Obviously, legal scholars and judges are also involved in the second dimension. Determining the law, particularly when the law is under-specified, intrinsically requires meta-level normative reasoning.

[12] Meta-ethics, in turn, addresses questions regarding the nature and construction of ethical claims.

[13] Edmond Awad et al. "The Moral Machine Experiment". In: *Nature* 563.7729 (2018), pp. 59–64.

Socially constructed morals Starting from their own etymology ("mores"), morals also align with societal structures, such as *niches*, *strata*, classes, communities, the "People". Behaving "well" can serve as a source or signal of good reputation within one group, but not necessarily in another—consider how charity events organized by upper social classes are considered by lower social classes, alongside the codes of honor in mafia or gang culture and their effects to society at large. The point here is that approval within a specific group does not supply a sufficient ethical justification for its conduct.

Morals as applied ethics Another perspective, consequent to how we defined ethics, suggests to see morals as a set of acceptable and unacceptable behaviors derived from explicit ethical principles—essentially, the main purpose of ancient Greek philosophical schools: guide someone on (deciding) how to live.[14]

Materially constructed morals None of the preceding accounts fully explain *why* specific social phenomena—such as slavery—exist in certain historical moments but not in others. More broadly, why are practices that benefit some at the expense of others accepted as normal, or even defended as right? A *materialist* perspective addresses this directly: slavery was accepted and defended insofar as it sustained the material power of ruling classes. Moral standards are socio-economically and historically conditioned, never formed outside the arrangements through which people live and exercise power. Evidently, this does not reduce every moral commitment to raw economic interest; rather, it prompts us to examine the structural conditions under which certain commitments become authoritative while others are marginalized. For AI, the corresponding question is: *whose interests are sustained by the standards embedded in a system, and who holds the power to define those standards?*

Evidently, these perspectives address distinct, at times overlapping, questions. Ultimately, explaining how a moral belief historically emerged does not resolve the question whether that belief is justified.

4 Norms and values

A useful distinction can be made between statements about **right/wrong** conduct and those about **good/bad** conduct. Traditionally, philosophy divides ethical inquiry into two primary perspectives closely related to this distinction: **deontology**, centered on the study of norms (typically expressed using deontic vocabulary: obligations, prohibitions, and permissions), and **axiology**, centered on the study of values. Usually, the former evaluates action from a procedural point of view (*performance* or non-performance), whereas the latter evaluates *outcomes*, *states of affairs*, or *character traits*, examining action from a substantive point of view, often abstracting away its procedural dimension. Yet, one should not conclude that deontological ethics holds a monopoly on defining what is right and wrong; as we will see, one can also adopt a value-based perspective to evaluate right and wrong actions (e.g., the right behavior is that which maximizes total value).

Exercise 4. Guardrails vs. Optimization

- *What comes first? Do valuable outcomes determine what we ought to do? Do obligations constrain which outcomes we may legitimately pursue? Can one framework exist independently of the other?*

[14] Evidently, knowing principles and applying them in practice are two very different things. Seneca, a Roman Stoic philosopher whose writings remain widely read, did not really align his life with his own teachings.

4.1 On the distinct ways to investigate value

4.1.1 Value in philosophy

Philosophy has long debated the distinction between things that are good for their own sake (*final value*) and things that are good for achieving something else (*instrumental value*).[15] Pleasure, the absence of pain, and, more generally, the satisfaction of basic needs or desires are classic candidates for the first category. Money is a familiar example of the second, because it is valued primarily for what it enables. Friendship generally belongs to both: it is worth as an end in itself, but also as a means (e.g. mutual support).

4.1.2 Value in economics

Because of its centrality to the determination of price, economics has proposed different theories on how to measure the value of a given entity (whether an artifact or a process):

- **Labor** or **production**: how much it costs to produce that entity.
- **Utility** or **use**: how much practical benefit or satisfaction it brings.
- **Exchange**: how much others are willing to pay for it on the market.

From a prescriptive point of view, the connection of price with moral dimensions echoes historical debates on "just price" (generally in contrast to market price) across religious and political philosophy. From a descriptive point of view, financial modelers typically analyze markets by distinguishing between *fundamentalists* (who maintain a grounded, or at least stable, estimate of an asset's worth, generally rooted in intrinsic or use theories of value) and *chartists* (who follow market dynamics as expressed through "charts", thereby aligning pure exchange value theory).

Production cost, practical utility, and exchange price frequently diverge, even dramatically. Blockchain technology offers a clear illustration of this divergence. Proof-of-work mining compensates computational effort directly, anchoring value in *production cost* (the "labor" of performing cryptographic calculations). As an operational tool, blockchain is evaluated by its *use value*, the actual improvement of efficiency or transparency it introduces into business processes (though whether this specific technology yields net positive utility remains debated). Non-Fungible Tokens (NFTs), that is, unique cryptographic signatures linked to digital assets, exemplify instead pure *exchange value*. Requiring minimal labour (e.g. copying a digital asset on the blockchain) and offering negligible direct utility apart from signaling, their valuation relies almost entirely on artificial scarcity and market speculation surrounding cryptographic proof of ownership.

4.1.3 Value in psychology

Values are clearly not only an economic matter. In general, they manifest as rather abstract concepts—consider, for instance, "honesty" or "integrity". Yet, because people use values to interpret the world and consequently talk through and about them, such expressions of value can, in principle, be empirically captured. In psychology, various frameworks exist to categorize fundamental human values. The most prominent of these is Schwartz's Theory of Basic Human Values, structured around two primary macro axes: *openness to change vs. conservation* and *self-transcendence vs. self-enhancement*. Connection holds with economic values, e.g. self-enhancement aligns with status signaling, and therefore underpins exchange value.

Exercise 5. Which values do AI agents have?

- Several recent papers use psychological or political value frameworks to survey LLM-based chatbots. *What are these evaluations actually capturing?* Does a classification of generated

[15] A related distinction is whether value depends on the entity's own properties (*intrinsic value*) or on their relationship with other entities and context (*extrinsic value*).

text automatically reveal stable commitments, underlying motivations, or an enduring moral identity?

- Apply the economic perspectives on value (labor/production, utility/use, exchange) to LLM-based chatbots. *Which of these components currently dominates the offer price of AI services, and what drives their demand?*

4.1.4 Value in law

Law, by its very nature, is formulated in general terms; it is intended to be *generally* applicable, including under circumstances different from those that motivated its initial adoption. For standard, common cases, legal norms typically provide sufficient guidance. However, *hard cases* inevitably arise in which interpretation requires *balancing* opposing interests and conflicting norms—the prototypical example being the tension between environmental protection and economic development. Judges and administrative authorities navigate these cases through legally structured reasoning rather than by invoking personal preferences. The need for normative anchors to support and evaluate this argumentative process explains why legislation, particularly in constitutional contexts, explicitly refers to values. The Treaty of the European Union (TEU), for instance, explicitly grounds in Article 2 to founding values such as human dignity, freedom, democracy, equality, the rule of law, and respect for human rights, including minority rights. Yet, declaring a value does not, on its own, guarantee its effective application.

5 Beyond *mores*: law and ethics

Social norms are generally described as *informal* norms, as they lack a formalized system for their creation, enactment, and adjudication. Legal systems, by contrast, emerged around and operate through a process of formalization, in which rules are articulated in written artifacts and proceduralized mechanisms are put into action to apply, modify, or enforce them. From this perspective, one may view legal norms as a specific subset of the broader normative system relevant to a community—with the important caveat that a community may not practically adopt or adhere to certain legal norms even if it is formally addressed by them. Actually, legal norms typically aim to guide people, and, by extension, their social norms into new territories. Legal norms thus fulfill the prescriptive dimension, but they do not necessarily fulfill the descriptive one. This imbalance captures precisely the governance function of law.

5.1 When laws are written

Legal scholars have made clear that a fundamental distinction exists between a state governed by the substantive *rule of law* and a state that merely operates through laws (*rule by law*). Nazi Germany and, more generally, totalitarian regimes typically maintain a functioning legal machinery; yet they fail to fulfill the substantive requirements of fundamental rights. This distinction generalizes beyond such extreme cases: written rules alone do not establish how power is constrained, how norms are applied in practice, or how decisions can be meaningfully challenged by people.

For AI developers, this serves as a critical warning against equating the mere existence of rules or constraints with adequate governance. Essential questions inevitably remain: *Who interprets these rules, and through which methods? Who enforces them? Who can contest their application, and through which procedures?*

5.2 When laws are not there

When laws are absent, or are perceived as inadequate to address a novel domain of societal concerns, ethics is often called upon to fill the normative gap. A typical example was the creation

of bioethical committees when genetics made great advances in the last 1990s. Much of the contemporary AI ethics discourse also originated from this perspective. However, scholars have also argued that many issues associated with AI already fall within existing legal frameworks, such as *product liability* (e.g., if a device causes damage or harm to a user, its manufacturer is held liable). Although the opaque, non-deterministic nature of modern AI challenges traditional notions of causation—and has made clear the presence of a *responsibility gap*—yet it is difficult to argue that who creates and deploys such systems has limited responsibility. On the contrary, several scholars have pointed to an ongoing phenomenon of "ethics-washing": centering the discourse primarily on which rules should govern AI (typically the domain of ethicists) and how to technically implement these guardrails (typically the domain of AI engineers) distracts attention from the more pressing and immediate issue of legal liability.

6 Core components of ethical frameworks

The different ethical frameworks proposed and discussed so far can be seen as being characterized by constructs expressing different forms of normativity, possibly conflicting.

Deontic commitment *You perform an action because it is due.* Duties, permissions, and prohibitions tell you what you must, may, or must not do. Such a deontological perspective on regulating behaviour relies on deontic commitments to establish behavioural boundaries. A prohibition is not simply an outcome to trade off against the expected benefit of breaking it.

Consequentialism *You assess actions through their consequences.* Said differently, you pursue an end and perform the action that enables you to achieve it. This connects this construct with *instrumental reasoning*. Typically, outcomes are evaluated in terms of values, making consequentialism the primary domain of axiology. Yet, one may express reaching a certain outcome as a duty as well. The difference lies in how side effects (which may have relevant consequences) are treated within the decision mechanism.

Virtue ethics *You behave as a virtuous person would.* Exemplary role models help orient conduct, shifting attention toward the kind of person—or professional—you are trying to become. The point is not merely to imitate a person's visible behavior, but to cultivate judgment about what a situation calls for. This typically includes recognizing which norms have priority and internalizing a balance between different outcomes.

Contractualism *You consider principles of conduct in relation to the terms on which people live together.* This often occurs through an unspoken social contract between members of a society (or, more generally, within a community in which you participate by following or endorsing its norms).

Discourse ethics *You consider whether your conduct can be justified in argumentation to those affected by (most notably suffering) its consequences.* This construct brings attention to participation, reasons, and the conditions under which the acceptance of regulation is formed.

Care ethics *You assess actions through the lens of interpersonal relational connections, focusing on taking responsibility for specific, vulnerable, or dependent others.* This construct centers on maintaining, repairing, and strengthening interpersonal networks to support those who are in need.

You may have noticed various relations connecting these constructs. Deontic and consequentialist frameworks are complementary, focusing either on a reference performance (expressed through norms) or on evaluating all outcomes of that conduct (by means of values). Virtue ethics composes normative or value-laden directives, along with priority or aggregation mechanisms, resulting in a reusable behavioral role that may span across domains. Contractualism, discourse ethics, and care ethics operate at a higher meta-level, addressing how collective norms and values interface with individual ones: the first examines how the regulated entity relates to the regulating entity; the second considers how the regulating entity grounds its evaluations back in the experiences of the regulated; and the third considers the whole network of relationships constituting the collective entity (viewed here as a multiplicity rather than a unified whole) and operates on its most sensitive internal threads.

6.1 Consequentialism, Utilitarianism, and Optimization

Utilitarianism is a specific variation of an axiological view on consequentialism—particularly influential in economics, and subsequently in AI—that evaluates outcomes in terms of value, expressed in terms of *utility* functions. Optimization is thus the mathematical technique backing utilitarian views on decision-making, used to select actions or policies to maximize utility (or a functionally equivalent objective function), such as e.g. an aggregated reward (or value) function in reinforcement learning or accuracy in a classification task. In this sense, contemporary AI is consequentialist by design, as its methods are typically organized around achieving specific, narrow outcomes. Even Large Language Models are fundamentally constructed to perform well in a text completion task.

From a general point of view, however, if the core policy of an agent reflects its operational identity (in the cybernetic sense of what characterizes the system in coupling with its environment), optimization operates as a meta-identity: a meta-level orientation that drives updates to that policy. The Machiavellian motto "the end justifies the means" is a provocative shorthand for a radical perspective on consequentialism. When only an outcome is valued (or is orders of magnitude more important than all else), everything leading up to it is treated as secondary; the system can transform into anything in order to reach that objective.

Exercise 6. Critical questions on optimization

- *Is optimizing an objective adequate for an agent to behave ethically?*
- *Who selects the objective? Whose interests does it express? Are there means which remain strictly unavailable?*

7 Two exercises of monitoring design

7.1 Where to put the speed cameras?

In your city, the number of car accidents is rapidly increasing. You are asked to find who is exceeding the speed limit, so that violators can be punished.

- You have only a limited number of speed cameras already deployed across the city. *Where should you move a number of them?*

Suppose you have 3 choices: (1) in areas where people usually exceed speed limits; (2) in areas where people usually do not exceed speed limits; (3) somewhere else.

If you replied (1) that's what most people would do. This is because putting more cameras where you expect violations seems the most *efficient* (you use more of the cameras you have where they are needed more) and *effective* choice (you find more violators). But let us look at the

data you will then collect. Not unexpectedly, you will observe more violations, as you look more intensively where more violations occur. If those observations justify placing still more cameras there, the monitoring strategy can reinforce its starting assumptions. Meanwhile, violations elsewhere will become less and less visible. This phenomenon can be seen as an informational variation of *self-fulfilling prophecy*: if we look for what we expect only where we expect it, then we may only see what we expect. This blindness is consequent to a blind optimization attitude! More precisely, it is a case of selective observation and a self-reinforcing feedback loop. For speed cameras, we do not expect that additional monitoring will cause additional violation; in certain cases, this may happen.

Obviously, going systematically for (2) or (3) does not seem right too, as we would miss the opportunity to reinforce the law where reinforcement is needed.

In short, blind optimization and no optimizations appear to be both wrong. What we acknowledge here is that there is a trade-off that needs to be decided, and further considerations are needed (e.g. disruption caused by additional monitoring, velocity of adaptation by violators, effects of violation).

Finally, there is another (not less important) question too. The problem began with increasing accidents, but the assigned task became detecting speeding for punishment. Maximizing detected violations is not necessarily the same as reducing accidents. Going one level up can definitively help appreciating the horizons of problems that exist related to the one we were asked to solve.

7.2 Catching synthetic drug production

A second exercise comes from an actual Master IS project. Students were asked to help the police in identifying venues of synthetic drug production. The project description suggested a pattern: barns were rented for a few months, then abandoned, with chemical residuals disposed of in canals. It also suggested that economic pressure on agriculture might make barn owners less demanding or attentive towards prospective tenants.

The initial idea of a group of students I supervised was: "Let us build a risk indicator using all economical information we possess (at a very high granularity): if an area is becoming poorer, we may expect barns there to be rented for drug production."

- *What is wrong with an individual risk indicator based on socio-economic indicators?*

At a fundamental level, the proposed indicator shifts attention from the addressed behaviour (which is performed by individual actors) to general socio-economic conditions (which are expressed by a population). In short, increasing poverty in an area becomes a basis for suspicion, even though it is not evidence that a particular person or barn is involved in drug production. Targeting inspection merely on that basis can also reproduce the selective-observation problem from the speed-camera example. Additionally, in this case we are sure that sending the police for control in households that are having financial issues will increase stress to people possibly already in distress, possibly traumatic effects on minors, time taken from economic activity, and possibly further poverty. The self-fulfilling prophecy risks to be material, not only informational.

7.3 Circumstantial/contextual vs direct behavioural evidence

One way to reduce these issues is to focus strongly on evidence related to the *modus operandi*: the behaviour being addressed. Avoid, as much as possible, relying on co-occurring properties such as socio-economic features or more in general *circumstantial* evidence provided by contextual proxies.[16] Ask yourself: *am I considering behavioural features characteristic of the conduct I am monitoring?* Circumstantial evidence is not illegitimate in itself, but it should only furnish support to direct, closely connected evidence of conduct.

[16] In law, "circumstantial evidence" has a slightly different meaning. It may be associated to behaviour, as e.g. foot traces on the crime scene, but require further inference to connect to the specific person.

Indeed, returning to the synthetic drug production case, what the students did to readdress the project was to focus on the monitored conduct, whose main components were the renting of barns and the throwing of chemical residuals in canals. Building on these elements, they then assumed that disposal occurred neither immediately nearby the barn, nor too far away, to reduce risk to be caught by the police. Triangulating public data of barn available for rent and water analysis from nearby canals, they found potential suspect locations, a few of which were confirmed through public news reports.

Exercise 7. Border control risk indicator

- *What do you think of a "terrorism-risk" indicator in border control based on a picture?*

Exercise 8. Elaborating with ethical constructs

Consider the synthetic drug production scenario. Examine the proposed monitoring and enforcement measures through the various ethical perspectives presented above. What directives would each suggest? You can take the following critical questions as support for your elaboration.

- **Deontic**: *Which are the behavioural oughts (prohibitions, obligations, possibly liberties)? Which second-order oughts are considered to deal with violations? (e.g. what the police should do)*
- **Consequentialist**: *Which outcomes should be pursued or which should be avoided? What intended and unintended consequences might different measures produce, for whom, and with what uncertainty?*
- **Virtue ethics**: *Which trade-off of values emerge from the different conducts? What is the best balance for you and why?*

Now, assuming you have a normative proposal, shift attention towards the justification of these measures, the conditions of participation, and the relationships they affect:

- **Contractualism**: *What is the relationship between the various stakeholder and the overall social gathering? Would the people involved justify your proposal? Which objections could they reasonably raise? Are the proposed normative actions going to modify this relationship?*
- **Discourse ethics**: *Have you procedurally given opportunity to stakeholders (particularly the most suffering ones) to question the proposal?*
- **Care ethics**: *What is the current relational status between stakeholders (in terms of dependency, responsibility, trust)? What would happen to the social fabric if the proposed normative action is applied? Which responsibilities of care follow from this?*

8 A critical perspective on algorithmic fairness

8.1 Bias as a technical term and as a social concern

I was personally trained as an electronic engineer, and I learnt to see *bias* from a purely technical point of view: a current, a voltage, an offset or a deviation structurally observed in operating conditions. Now, in probabilistic terms, if the uniform case is deemed to be a baseline or null hypothesis, a non-uniform distribution of outcomes would be a deviation; from this perspective, any meaningful information requires some form of "bias"; otherwise, we would be in a condition of *maximum entropy*: pure noise. In statistics, bias is generally used to capture estimation error. In the machine learning pipeline the term has gained still other meanings, and several types of

bias have been identified (e.g., world bias, representation bias, measurement bias, aggregation bias, evaluation bias).

In common language, however, and even more so in recent times, the term "bias" is rather used to denote *social discrimination*: preferring someone over someone else. Often the concern is one of *negative discrimination*—that is, disfavouring or punishing people—but differentiation can also be used to favour or to assist. This is indeed the idea behind *affirmative action* initiatives: active efforts to improve opportunities for groups historically subjected to discrimination. Both negative and positive discrimination require differentiating between situations or groups. So, differentiation is not unjust in itself; to evaluate it one needs to examine its grounds and its further consequences.[17]

8.2 Statistical parity is inadequate to capture the fairness problem

In data-centred contexts, questions of social bias are often operationalized through *statistical comparisons between groups*. In this framing, Algorithmic fairness research have introduced various measures, such as *demographic parity* and *equalized odds*, which have been also proven to be incompatible. A system cannot be "fair" according to all definitions at the same time. Which one should we choose then? It essentially depends on the decision context—for instance, on the impact of a false negative compared with a false positive. A practical heuristics is proposed by the AEQUITAS fairness tree.[18]

The position I want to foreground here is however that a metric cannot, on its own, decide what fairness ought to mean; its choice is already connected to judgments about relevant harms, benefits, and responsibilities, which are about extra-algorithmic concerns. I will in the following focus illustratively on two further problems about applying across metric-based approaches.

8.2.1 Parity between whom?

Statistical measures rely on identifying groups; but which groups should we consider? Commonly, comparisons use protected or sensitive dimensions, such as sex or gender, race or ethnicity. Yet differences can appear at intersections. Consider for instance:

- women vs men;
- black women vs black men;
- low-class black women vs low-class black men.

It is not evident what is the right granularity to apply for a given problem. On the one hand, broad comparisons can hide subgroup differences. On the other, increasing granularity raises questions about sample sizes, uncertainty, and the relevance of the resulting categories. *Intersectionality*, as a branch of socio-technical research, is not just a request to create more categories to a dataset; it also directs attention to how social positions and structures of disadvantage interact. The fundamental question is what are the relevant categories to look at.

8.2.2 Parity about what?

Many algorithmic fairness approaches focus on the outcome of a decision predicted by a machine learning model, which is typically binary (e.g., granting credit or hiring someone). But is this what fairness truly means to humans? Philosophy has long debated which dimensions are relevant for evaluating and comparing people's situations. These can be summarized as *need*, *merit*, *fitness*, *resources* (concerning the decision subject); *access* and *treatment* (concerning the interactional mechanisms between the subject and the decision-maker); *resources* required to achieve a decision

[17] Interestingly, for a European participating in an American conference, reading a field 'race' on a registration form could be perceived as outrageous, if not illegal, although it may have been motivated by affirmative action purposes or even just to study possibly discriminatory social phenomena.

[18] http://datasciencepublicpolicy.org/our-work/tools-guides/aequitas/

(spent by the decision-maker); the *outcome* of the decision; its resulting *impact* on the decision subject and *distributive effects* (concerning ecological dimensions). It remains unclear whether there may be further dimensions.

What is clear, however, is that a decision-level metric centered on outcome selects only a fraction of that larger question. Similar treatment or outcomes produced by a model do not guarantee similar opportunities, burdens, or longer-term impacts. By focusing exclusively on the decision point, algorithmic fairness adopts a static, short-sighted consequentialist perspective. Improving a local score most plausibly will not yield fairer results overall, particularly as the wider social ecology shifts behaviours, opportunities, incentives, and cumulative effects emerge.

9 Preventing and identifying ethical issues

At a fundamental level, preventing ethical failures requires confronting a systemic challenge: mainstream technological, organizational, and societal practices today treat problems primarily as objective functions to be optimized. As a developer, you have at least three lines of counteraction to draw upon.

9.1 Define behavioural boundaries

One approach is to define clear boundaries through deontic directives. For instance:

- You are not allowed to clone humans. No matter what.
- You have to use conduct-related evidence for judgments. No matter what.

These statements are examples of unconditional commitments; they should not be taken as statements pretending to be universally applicable law, nor do they settle all processes supporting their application. The practical question is indeed how boundaries like these become effective. For AI developers, how they enter design requirements, constrain operation, and survive pressures to optimize performance.

9.2 Take on professional commitments

Another approach invokes virtue ethics: behave as a virtuous professional would do. Relevant examples are the *Hippocratic Oath* for doctors,[19] but also the (unfortunately far less known) *Archimedean Oath* for engineers.[20] From a public perspective, these commitments form the basis for professional deontology and public shared expectations about a profession. Evidently, the existence of an oath does not establish that people will follow it, or will change professional practice. Yet, it may increase social pressure, if promoted adequately.

9.3 Look the system from the outside too!

Taking a designer stance naturally directs our attention towards an internal, prescriptive view of a system: *what it has to do*. But looking at systems externally, independently of the design phase, is crucially important too.

First, the reasons why a system behaves in certain ways may be inaccessible or unintelligible. AI systems have become increasingly opaque to us. For instance, we do not easily understand the processes underlying activations in artificial neural networks. Yes, there is a rich stream of research working on this frontier, such as *mechanistic interpretability* and, more generally, *explainable AI* efforts. The point here is that we can take a dual perspective on the problem, assuming that we cannot access the system's internal workings. Applying the notion of *encircling*, borrowed from security studies (e.g. *What is this building used for?* or *What is inside this building?*), is a

[19] https://en.wikipedia.org/wiki/Hippocratic_Oath
[20] https://sefi2024.eu/conference/archimedean-oath/

relevant option for AI research too. Applying this approach, rather than focusing on the system itself, we study the activities which surround it.[21]

Second, even when systems are accessible, what they are designed to do may differ from what they do in practice. Consider, for instance, the phenomena emerging from contemporary recommender systems. We need to explicitly distinguish *motivating* aspects—the commitments that guide or justify design—from *revealed* aspects, visible through operation and effects. We also need to consider how those effects are experienced by affected people.

To evaluate artefacts and systems independently of their official commitments and requirements opens at least two complementary lines of empirical investigation related to ethics: which normative orientations systems express in practice (*empirical ethics*), and how people interpret, accept, or contest those systems through their own moral and ethical frameworks (*descriptive ethics*).

9.4 Possible blind spots

The above aims to put to the foreground that several dimensions are relevant while designing a system which is going to have a societal deployment. As AI developers, we need to agree that intelligent artificial systems, almost intrinsically, because they modify the cognitive abilities of humans and the arrangements through which people act, risk having wider effects on the social fabric.

Dimension	**Core question**
Prescriptive vs *Descriptive*	What should happen, and what happens in practice?
Internal vs *External*	What can be understood through design access, and what can be examined from outside?
Motivating vs *Revealed*	Which commitments are stated, and which orientations appear in operation?
Non-technical vs *Technical*	Which questions require social, institutional, or ethical analysis, and which require technical investigation?

These dimensions are related, but not interchangeable: access to code does not automatically reveal social effects, external observation does not automatically explain design intent. It is also particularly relevant to consider how these investigations inform each other. This list of questions offers a first screening to identify possible blind spots in your work.

10 Responsibility and agency

We have looked at concepts related to ethics, examples of problematic applications, and critical questions about algorithmic fairness. Many relevant points remain outside this crash guide, including privacy, consent, participation, accessibility, labour conditions, environmental impacts, avenues for contesting decisions, and post-deployment governance.

The examples I have shown were meant to illustrate that there exists no one-fits-all heuristic. There is no context-free "solution" (or even "problem definition") that makes further judgment unnecessary. Yet discussing ethics assists us in forming adequate reasons to justify why we accept—or do not accept—an algorithm working in a certain way. This is a matter of individual and collective responsibilities.

[21] Bernhard Rieder, Georgy Gordon, and Giovanni Sileno. "Mapping Value(s) in AI: Methodological Directions for Examining Normativity in Complex Technical Systems". In: *Sociologica* 16.3 (2023), pp. 51–83.

10.1 With power comes responsibility

As an AI designer or developer, you can draw on prescriptive ethics to articulate the specific norms and values your system should embody. The familiar system administrator's warning in Unix/Linux-based systems has popularized this directive in technical domains: *Respect the privacy of others. Think before you type. With great power comes great responsibility.* Technical capability is not, by itself, permission. The ability to build or deploy something brings along questions about the grounds on which you should do so.

10.2 Your actions have consequences

As an AI designer or developer, and as a social actor, you can consider empirical and descriptive approaches to ethics to assess which norms and values systems realize in practice, and what is expected by the social system in which it will be deployed. A stated commitment is not enough. Look at how the system operates, which burdens it creates, and how its consequences are experienced by those affected.

10.3 Protect your agency: do not be "instrumental"

Yet, we need also to be clear on a complementary point: *individual responsibility* is only half of the story. Developers are only one part of the chain. Addressing individual behaviour alone may cast shame on people who, for material reasons, have limited room to act. It is no coincidence that, throughout history, dominant social groups have routinely framed lower socioeconomic classes as immoral or amoral. This observation obviously does not erase individual responsibility. It crucially asks us to examine how responsibility and power are distributed *before* talking about morality, and to look at structural concerns as the target for interventions.

Exercise 9. Distribution of responsibility in production

In your view, *who bears responsibility for weapon production?*

- The designer who conceives weapons?
- The worker who materially produces them?
- The company that sells them?
- The country that allows their production and sale?

If this exercise was sufficiently clarifying, it should illustrate why it is important, particularly as an AI developer, to participate in steering those who have discursive, political, and economic power in society. Communities of practice, associations, NGOs, worker organizations, and labour unions are possible directions for doing so, because *collective action* can address conditions that isolated developers cannot change on their own. Protecting your agency means more than making personal ethical choices. It crucially concerns institutions by which choices become possible and effective.